

AI WasteWise

AI-Powered Waste Segregation & Sustainability Assistant

Presented by Devarapalli Sravani

CSE • Lakireddy Balireddy College of Engineering

1M1B AI for Sustainability Virtual Internship

Problem & Need

The Problem

Who Is Affected

Why AI?

People often struggle to identify the correct waste category and disposal method, leading to improper segregation.

Students, households, offices, campus staff and local communities.

AI can classify waste from text or images and provide quick, understandable disposal guidance.

How might we use AI to help people identify and properly segregate waste?

Goal: make responsible waste decisions easier at the point of disposal.

Design Thinking Approach

1

1 EMPATHIZE

Understand user confusion around waste disposal.

2

2 DEFINE

Frame the waste identification and segregation challenge.

3

3 IDEATE

Explore an AI assistant using text and image inputs.

4

4 PROTOTYPE

Build a web prototype connected to Gemini AI.

5

5 TEST & REFINE

Test common and uncertain waste examples and improve guidance.

SDG Alignment

Primary SDG — 12

Responsible Consumption and Production AI WasteWise supports better waste identification and segregation, encouraging more responsible consumption and disposal practices.

Primary focus: SDG 12

Secondary SDG — 11

Sustainable Cities and Communities Better segregation at source can support cleaner communities and more sustainable waste-management practices.

Proposed AI Solution



AI Architecture & Workflow

User Text / Image

Frontend

Node.js Backend

Gemini AI

Classification + Disposal Guidance

→

Working Prototype

Text Analysis

Image Analysis

User enters an item such as “plastic water bottle”. AI returns category, disposal guidance, confidence and a sustainability tip.

User uploads a waste image. The prototype sends the image to Gemini AI and displays the resulting classification and guidance.

Responsible AI

Fairness

Avoid assuming one disposal rule fits every context; improve coverage across waste examples.

Transparency

Show what the AI classified and provide a confidence/verification signal.

Privacy

Minimize unnecessary collection of personal information and handle uploaded inputs responsibly.

Ethical Use

Encourage safe, responsible disposal and make uncertainty visible rather than pretending certainty.

Expected Impact

Environmental

Technology

- Make waste guidance easier to understand • Help students and households make quick decisions • Build sustainability awareness

Impact described here is expected/planned; it is not a measured field-study result.

Social

- Encourage better segregation at source • Reduce avoidable mixing of waste • Promote more responsible disposal decisions

- Demonstrate multimodal AI for sustainability • Combine classification with decision support • Provide a foundation for future campus deployment

Conclusion & Future Scope

Conclusion

AI WasteWise demonstrates how generative AI can support everyday waste identification and segregation through a simple text/image interface. It connects AI classification with practical disposal guidance and responsible-AI considerations.

Thank You

Future Scope

- Improve image recognition across more waste types
- Add localized recycling/disposal information
- Add multilingual support
- Build a campus waste analytics dashboard
- Extend to mobile and smart waste systems

AI WasteWise • SDG 12 • Responsible AI for Sustainability